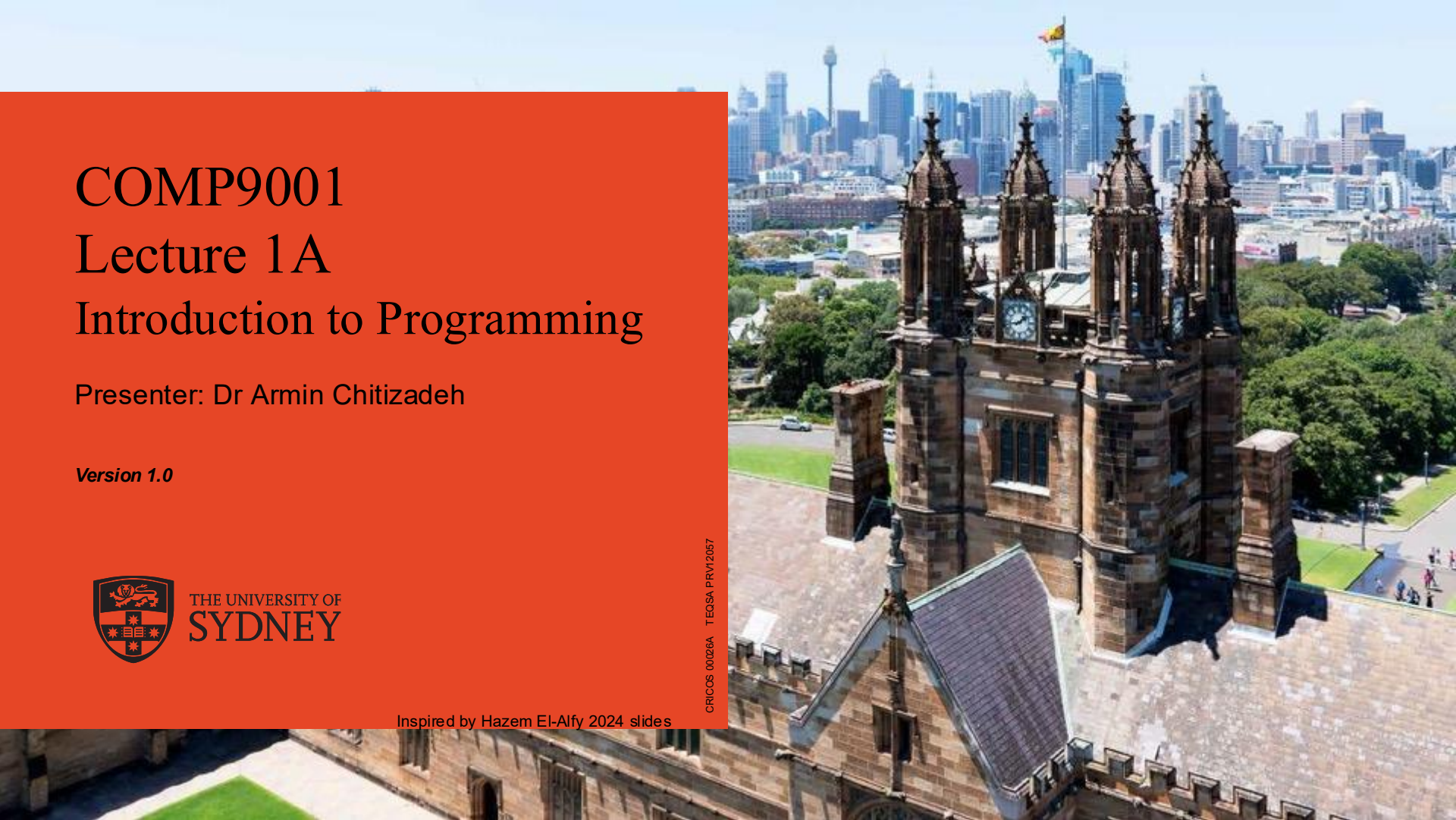
COMP9001
Lecture 1A
Introduction to Programming

Presenter: Dr Armin Chitizadeh

Version 1.0



CRICOS 00026A TEQSA PRV12057

Inspired by Hazem El-Alfy 2024 slides

We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

THE UNIVERSITY OF SYDNEY

Introduction to Programming

Introduction to Programming

- What is a program?
  - A set of instructions for a computer to execute in sequence to perform a certain task.

- What does a programmer do?
  - A programmer writes the instructions that become computer programs
  - They may write alone or together with a group of programmers.
    - Important for written program to be readable and understandable by other person
    - Also, it is important for a programmer to be able to explain their codes to others

Computer Programs and Recipes

- Like recipes, programs contain ingredients and instructions.

INGREDIENTS

INSTRUCTIONS

Hummus Recipe

3 cloves garlic
16 oz canned garbanzo beans
3 tbsp lemon juice
3 tbsp tahini
¼ tsp salt

1) Peel and chop garlic.
2) Open garbanzo can and drain the fluid.
3) In a blender, blend garlic.
4) Add all other ingredients and blend.

(c) 2018 Chand T. John

How Did You Get Into Today's Lecture?

- These are the series of actions that you have executed today for online students:
  1. Enter URL: https://canvas.sydney.edu.au
  2. Launch Zoom.
  3. Launch a web browser.
  4. Turn on the computer.
  5. Select 'Join'.
  6. Select the unit's Canvas site (COMP9001).

- These are the series of actions that you have executed today for in person:
  1. Get on the train
  2. Get off at Redfern
  3. Enter the lecture room
  4. Walk to lecture room
  5. Sit at the sit
  6. Open the door
  7. Find an empty seat

- Is the sequence of actions listed in the correct order to join?

Stop: Thoughts and Questions

- I signed up to learn about programming, not write instructions!
  Why is the lecture so hung up about the sequence of actions?

- Why don’t you tell me how to write my first program?

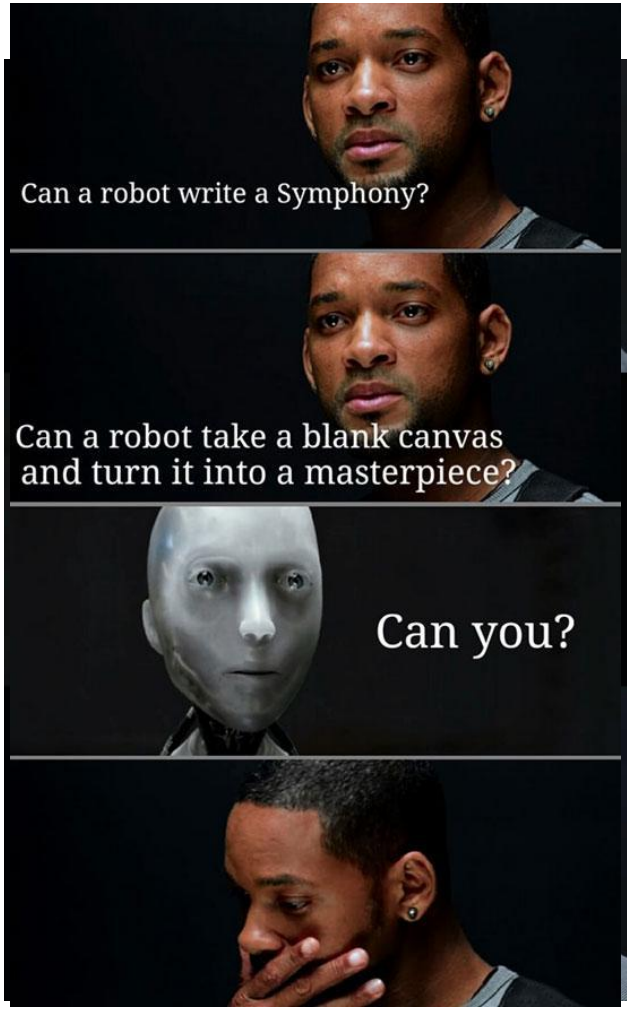
Reality of Programming

- A **program** is a set of instructions for a computer to execute in sequence to perform a certain task.

- The **programmer** writes the instructions that become computer programs.

- Can computers think on their own?

https://www.menti.com/alsx4fi391im


Can a robot write a Symphony?
Can a robot take a blank canvas
and turn it into a masterpiece?
Can you?

Understanding Computers

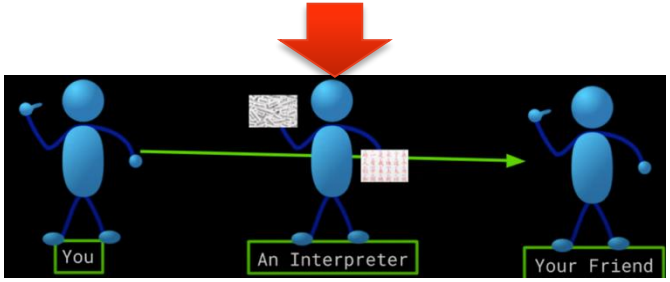
- Computer are precises and logical. They are good at doing repetitive tasks without complaints if and only if the instructions are clear and correct.

- The program or source code is a text file that contains the commands to be executed by the computer. The commands must follow the syntax of the language

- Programmers must be precises and logical when writing instructions. If they are not, programs would not work as expected and may even fail to compile.

What is Programming? Instruct a Friend

- You instruct your friend who doesn't understand your language, how to build a Lego set.
- Your friend lost the original instructions so can only build based on your commands.



Pictures from freeCodeCamp.org

Activity – Writing Instructions (10 mins)

- Write instructions for a fellow student to find their device's Operating System (OS)

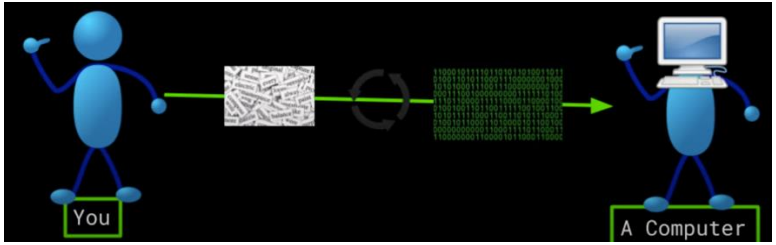
- Assumptions:
  - Student only has one device.
  - Student only understands English.
  - **Student does no know what their OS is and where to find it.**
  - URL to the Ed thread is given.
  - Student has access to the Ed discussion forum and permission to post.

Hints:

- Do an internet search about *Operating Systems* if you don't know what it is.

What is Programming? Computer Instructions

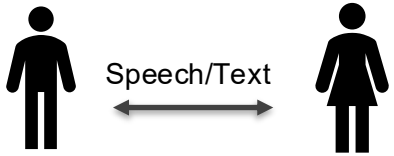
In order to talk to a computer, you must first give them instructions and then translate your English instructions to Binary.



But that's hard! You need programming languages for that

Giving Instructions to Humans vs. Computers

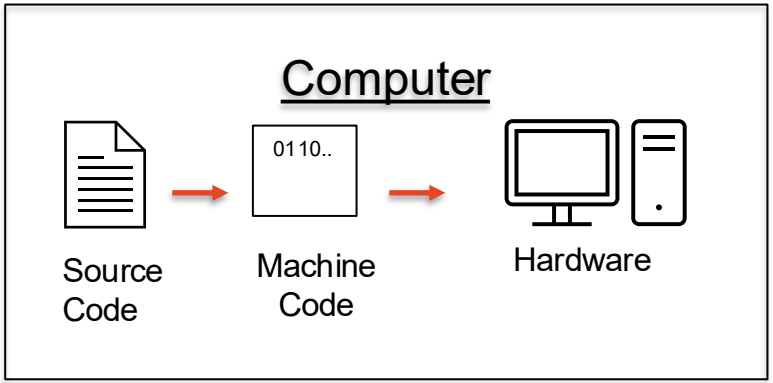
Human Language (HL)



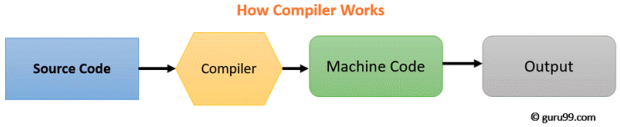
- Mandarin
- English
- Cantonese
- Farsi
- Hindi
- Arabic
- Turkish…

Programming Languages

Text or Graphics



Interpreter, Compiler



Compilers vs. Interpreter (part.1)

What is a Compiler?

A compiler is a computer program that transforms code written in a high-level programming language into the machine code. It is a program which translates the human-readable code to a language a computer processor understands (binary numbers 1 and 0 bits). The computer then processes the machine code to perform the corresponding tasks.

A compiler should comply with the syntax rule of that programming language in which it is written. However, the compiler is only a program and can not fix errors found in that program. So, if you make a mistake, you need to make changes in the syntax of your program. Otherwise, it will not compile.

Compilers vs. Interpreter (part.2)

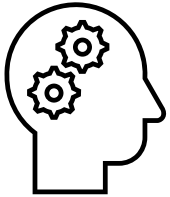
What is an Interpreter?

An interpreter is a computer program, which converts each high-level program statement, one by one, into machine code. This includes source code, pre-compiled code, and scripts.

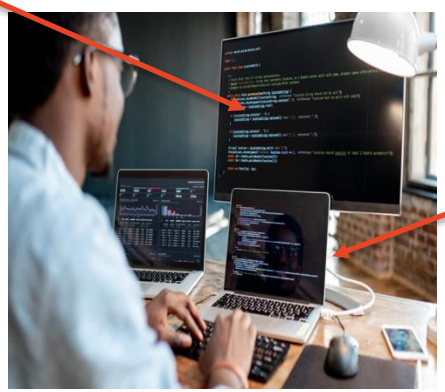
Both compiler and interpreters do the same job which is converting higher level programming language to machine code. However, a compiler will convert the entire source code into machine code (create an exe) before the program runs. Interpreters convert code into machine code while the program is running.

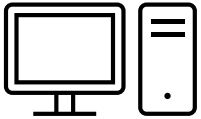
Learning to Code/Program

Knowledge
- Programming concepts
- Python Language



Affective (Behaviour)
- Work ethics
- Professional values





Psychomotor (Actions)
- Write, execute, debug and test programs.

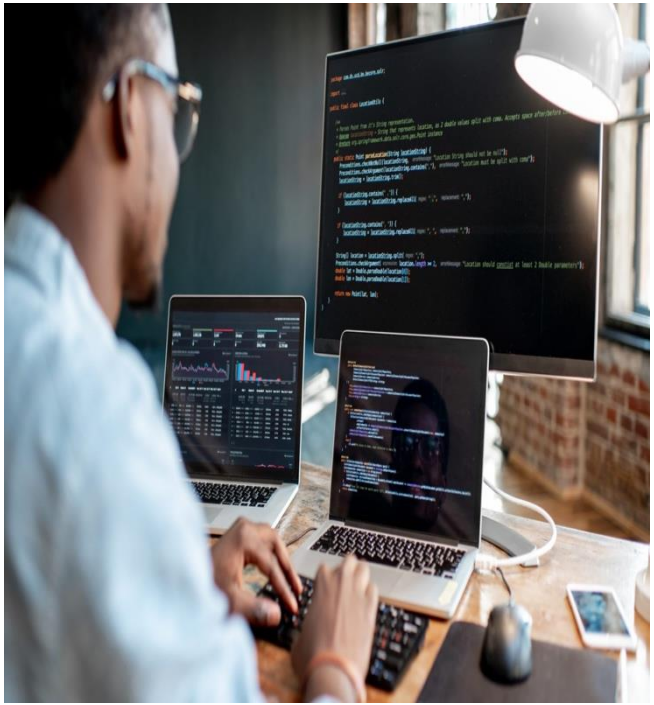
Learning to Code/Program

- Programs generally transform or process a given set of data into a desired output.

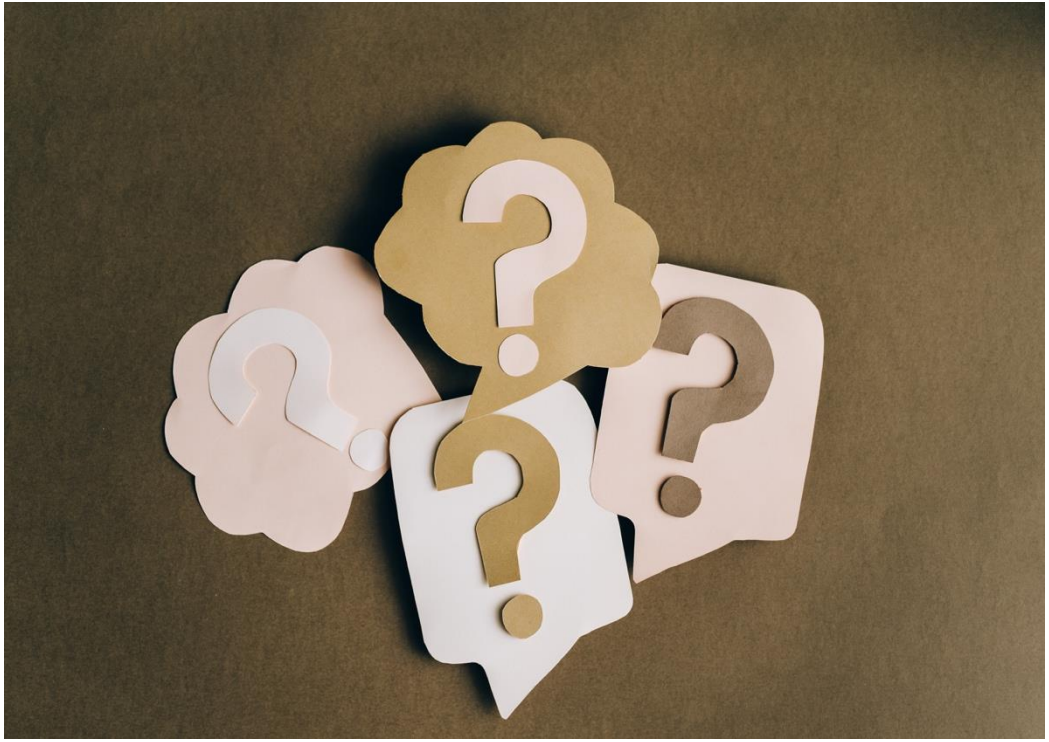
- Basic programming concepts:
  - Input
  - Output
  - Math calculations
  - Conditionals
  - Repetition

What Do I Need to Write a Program?

Components	Program
Tools	Text Editor + Terminal Pen and paper
Contents	Instructions, in code form, to do something or links to other programs.
Audience	Computer + Humans (other developers)
Language	Many languages. Different languages have different syntax.
Activities	Plan Code Test and Debug



Questions?



Summary of Part A of the Lecture

- A **program** is a set of instructions for a computer to execute in sequence to perform as certain task.

- The **programmer** write the instructions that become computer programs.

- For instructions to be executed correctly, they must be precise and logical.

- Initial steps to programming – understanding the problem and writing the algorithm in human language.

- Code is compiled or interpreted into machine language.

A Short Tour of Canvas and Edstem for this course

Break!

https://www.menti.com/aljih7byejc9

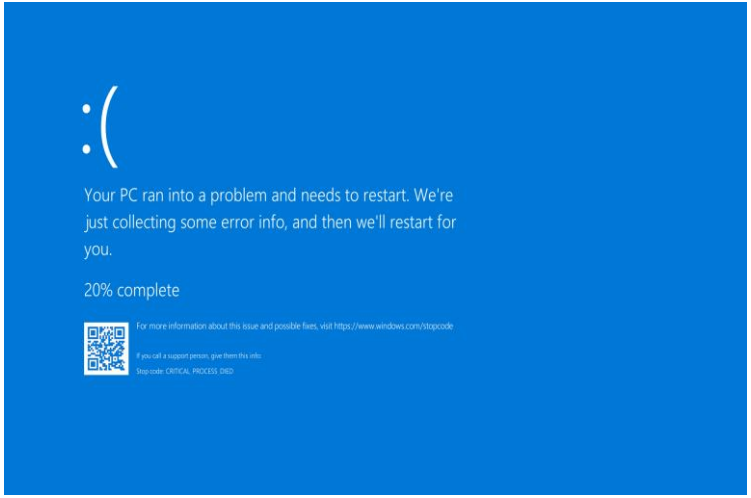
What Do you Expect to See When …

- A friend is happy?

- Your pet cat is hungry?

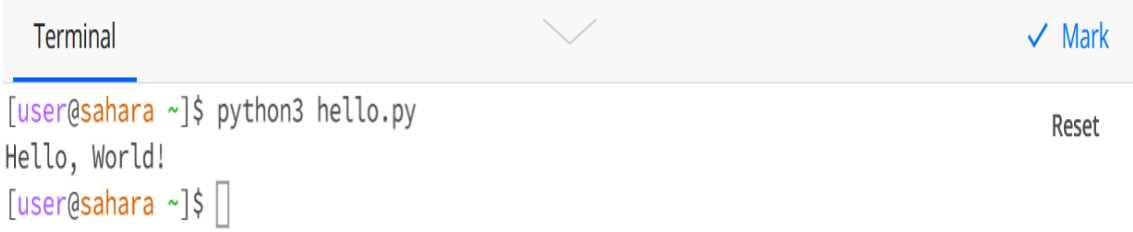
- Your computer is not responsive?

- Your program is working?



Your First Program

- Text output to the terminal
- Traditionally, a greeting from the perspective of the computer e.g. `Hello, World!`



Recap: What is a Program?

- A program is a set of instructions for a computer to execute in sequence to perform a certain task.

- Python is a high level language – close to human language but follows a special format (syntaxt)

- Three steps to code:
    1. Create .py extension file using Graphical User Interface (GUI) or Command Line Interface (CLI).
    2. Write code into the file.
    3. Run the file (the program).

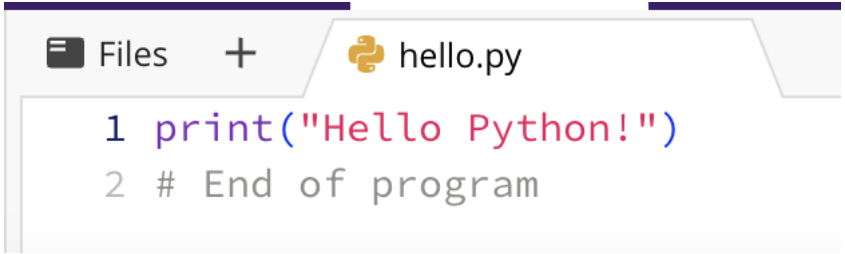
Your task:

Try out the three steps in the Ed lesson .

Step 2: Write Code Into File

Your task:

- Type both lines of text into the file hello.py
- Make sure you follow the image *exactly*



Step 1: Create File

- Command line interfaces – has learning curve but powerful

- Common actions that you do on the computer:
  - Directory (Windows user -> Folder)
    - Where am I: `pwd`
    - View contents: `ls`
    - Create new directory: `mkdir [name]`
    - Change directory: `cd [path]`
  - Working with files
    - Create file: `touch [file]`
    - Move file: `mv [file] [dest]`
    - Copy file: `cp [file] [newfile]`

Step 2: Write Code Into File Using VIM

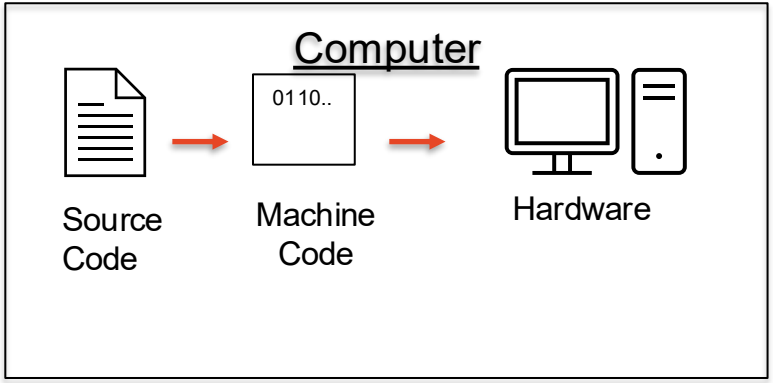
1. Open file in VIM.
2. In Normal mode (press Esc to switch to Normal mode), press I to switch to insert mode and type in the file.
3. Save file by going back to Normal mode with :w
4. Quit vim with :q

Step 3: Run the program

- The Python interpreter convert each line of text to machine code and executes it.
- Machine code is binary – 1's and 0's
- The compiler also converts the source code into executable codes.
- Challenge: Can you tell the difference between a compiler and an interpreter?

Terminal

```
[user@sahara ~]$ python3 hello.py
Hello, World!
[user@sahara ~]$
```



Mark

- Now that we have written our first program in the first lecture and have done the lab exercises, it’s time for us to go back and press that button in Ed and see what will happen

- Go to Ed
- Press the Mark button

Compiling a Python Program

- You can instruct the computer to compile the Python source code:

```
> python3 -m py_compile [file_name]
```

- This creates a `.pyc` file for your program which is in machine code (bytes)s

- `py_compile` is the compiler program

Text Output – `print()`

- A built-in function that is always available (with correct installation).

- What do you observe?

- Documentation: https://docs.python.org/3/library/functions.html#print

hello.py

```
1 print('Hello, World!')
2 # End of program
```

Terminal

```
[user@sahara ~]$ python3 hello.py
Hello, World!
[user@sahara ~]$
```

Python Interpreter

- You can run the Python interpreter, also known as python shell, by typing `python3` in the terminal.

- The interpreter instructs computer to execute codes and display result by returning output to the terminal immediately.

- To exit the interpreter:
    - Press CTR-Z on windows
    - Press CTRL-D on Linux
    - Type quit()

- An alternative to the python interpreter is the Interactive Python shell. Launch it by typing `ipython` in the terminal.

  Python Documentation: https://docs.python.org/3/tutorial/interpreter.html

Activity – Python Interpreter

- What would the following codes produce when executed?

```
print("#")
print('$$")
```

- Check your answers with the Python Interpreter.

First Program Reflection

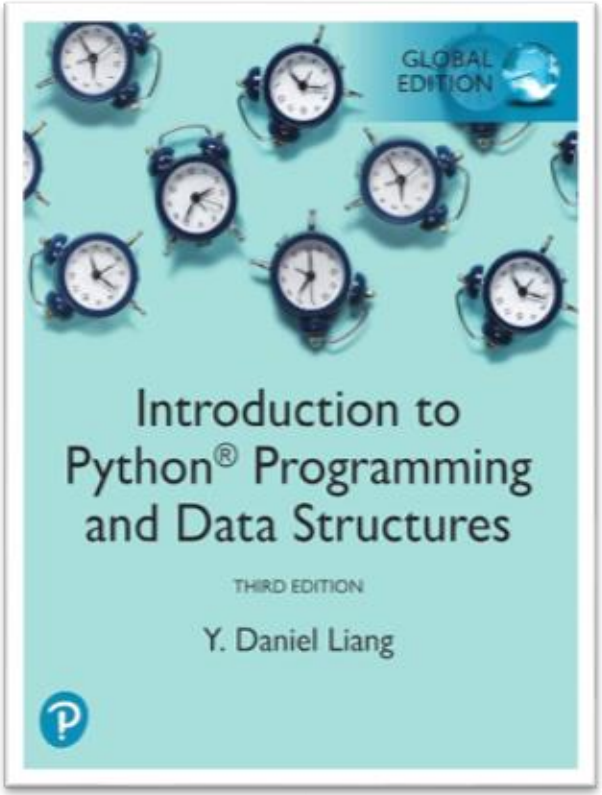
- Did you successfully write and run your first program?
- Which step did you find most confusing or difficult?
- Go to Ed

Summary – What Did We Do?

- Wrote and ran our first Python program.
- Introduce some tools for programming:
    - Interact using GUI/CLI
        - Navigation
        - Creating file
    - Exit text file within
        - VIM
        - Ed Workspace
    - Compile, debug and execute program
    - Troubleshoot and help
        - Experiment with Python Interpreter
        - Python documentation

Self-Study (if you wish)

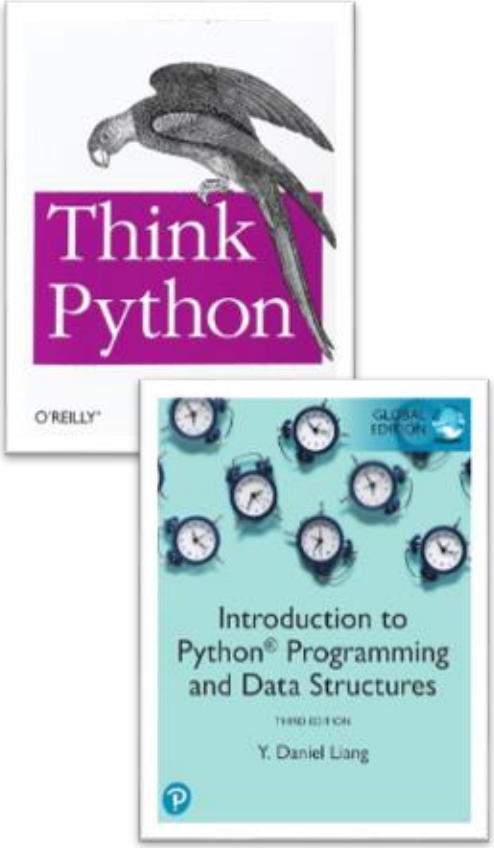
- **Read Chapter 1.** Lian, Y. D. (2022*). Introduction to Python Programming and Data Structures*. Pearson.



Reading (if you wish)

- Chapter 1. Downey, A. B. (2015). Think Python: How to Think Like a Computer Scientist (2e ed.). O'Reilly Media, Incorporated.

- Chapter 1.2 - 1.3, 1.6. Liang, Y. (2022). Introduction to Python Programming and Data Structures, Global Edition. Pearson Education Limited.



Assessments

- You need to explain your codes for assignment1 and assignment2

Updates

Fully anonymous feedback form for COMP9001 students – Fill out form